

WBOCRM - Requirement Traceability Matrix

Istanbul Arel University - Software Requirements Engineering HW5 | May 2026

This document traces every requirement from the READ specification through architecture, implementation, testing, and UI verification.

1. High-Level Requirement Coverage

Maps each high-level requirement from the READ specification to functional/non-functional sub-requirements and implementation status.

Req ID	Description	Sub-Requirements	Implementation	Status
FR-ST-01	Centralized lead management module	FR-LM-01..06, FR-SC-01..06	Lead CRUD + scoring + assignment	PASS
FR-ST-02	Customized Lead Scoring Algorithm	FR-SC-01, NFR-ST-10	scoringService.js (5-metric 0-100)	PASS
FR-ST-03	Sales Force Automation (SFA) tools	FR-LM-03..06, FR-ST-04	Pipeline stages + interaction logs	PASS
FR-ST-04	Marketing Automation modules	FR-SC-13	campaign_id field on Lead	PASS
FR-ST-05	Centralized Customer Support module	FR-ST-01..02, FR-ST-09..10	Ticket CRUD + optimistic locking	PASS
FR-ST-06	Real-time data visualization dashboards	FR-ST-11, FR-UC-12, FR-ST-14	DashboardPage + 2s auto-refresh	PASS
NFR-ST-01	GDPR and KVKK data protection	NFR-ST-07	PII masking + erasure endpoints	PASS
NFR-ST-02	Role-Based Access Control (RBAC)	FR-AUTH-01..04	authMiddleware + rbacMiddleware	PASS
NFR-ST-03	Cross-platform web-based frontend	Section 3.8	React SPA, min 360px viewport	PASS
NFR-ST-04	Low-latency RESTful API performance	All FR-* endpoints	Express REST API, parameterised SQL	PASS
NFR-ST-05	High usability for SME staff	Section 3.8	Responsive UI + NavBar + forms	PASS

2. Functional Requirement to Source Code Traceability

Maps each functional requirement to the specific source file(s) and function(s) that implement it.

Req ID	Description	Source File	Function	API Endpoint
FR-AUTH-01	JWT login	authController.js	login()	POST /api/auth/login
FR-AUTH-02	401 on invalid credentials	authController.js	login() - 401 branch	POST /api/auth/login
FR-AUTH-03	401 without token	authMiddleware.js	verifyToken()	All protected routes
FR-AUTH-04	403 insufficient role	rbacMiddleware.js	allowRoles()	Role-gated routes
FR-LM-01	Create lead	leadController.js	createLead()	POST /api/leads
FR-LM-02	Duplicate email check	leadController.js	createLead() - dup check	POST /api/leads
FR-LM-03	List leads sorted by score	leadController.js	getLeads()	GET /api/leads
FR-LM-04	Update lead	leadController.js	updateLead()	PUT /api/leads/:id
FR-LM-05	Delete lead	leadController.js	deleteLead()	DELETE /api/leads/:id
FR-LM-06	Interaction log notes	leadController.js	addLog(), getLogs()	POST/GET /api/leads/:id/logs
FR-SC-01	Priority score 0-100	scoringService.js	calculateScore()	Called at lead creation
FR-SC-06	Lead assignment	leadController.js	updateLead() - user_id	PUT /api/leads/:id
FR-SC-13	Campaign linking	leadController.js	createLead() - campaign_id	POST /api/leads
FR-ST-01	Create ticket	ticketController.js	createTicket()	POST /api/tickets
FR-ST-02	Update ticket status	ticketController.js	updateTicket()	PUT /api/tickets/:id
FR-ST-04	Interaction logging	leadController.js	addLog(), getLogs()	POST/GET /api/leads/:id/logs
FR-ST-09	Ticket list	ticketController.js	getTickets()	GET /api/tickets
FR-ST-10	Ticket priority tagging	ticketController.js	createTicket() - priority	POST /api/tickets
FR-ST-11	Active leads KPI	dashboardController.js	getDashboard()	GET /api/dashboard
FR-UC-02	Dynamic scoring	scoringService.js	calculateScore()	5-metric weighted
FR-UC-05	Pipeline stage update	leadController.js	updateLead() - pipeline_stage	PUT /api/leads/:id
FR-UC-12	Open tickets KPI	dashboardController.js	getDashboard()	GET /api/dashboard
FR-ST-14	Monthly revenue KPI	dashboardController.js	getDashboard()	GET /api/dashboard
FR-DOC-15	CSV export	leadController.js	exportCsv()	GET /api/leads/export/csv

3. Non-Functional Requirement to Source Code Traceability

Req ID	Description	Source File	Implementation	Scope
NFR-ST-01	Rapid lead entry (<3 clicks)	LeadForm.jsx	Modal form with auto-score	UI verified
NFR-ST-02	Cross-platform consistency	index.css + responsive layout	min-width: 360px	All pages
NFR-ST-03	Initiative Support Dashboard	DashboardPage.jsx	<5s load, 2s auto-refresh	GET /api/dashboard
NFR-ST-04	Core business uptime 99.5%	Render deployment	render.yaml blueprint	Infrastructure
NFR-ST-05	Concurrent data collision	ticketController.js	Optimistic locking (updated_at)	PUT /api/tickets/:id
NFR-ST-06	Modular release updates	Monorepo structure	backend/ + frontend/ separation	Architecture
NFR-ST-07	GDPR/KVKK PII masking	leadController.js	maskEmail(), maskName()	Support role masking
NFR-ST-07	Right-to-erasure	leadController.js, userController.js	erasePersonalData()	DELETE endpoints
NFR-ST-08	Bcrypt password hashing	authService.js	hashPassword() saltRounds=12	Registration
NFR-ST-09	Idle session timeout	AuthContext.jsx	30-min inactivity listeners	Frontend
NFR-ST-10	Algorithmic scoring speed	scoringService.js	calculateScore() <500ms	Lead creation
NFR-ST-12	Dashboard 2s auto-refresh	DashboardPage.jsx	setInterval(fetchData, 2000)	Frontend
NFR-ST-14	Offline ticket draft	TicketPage.jsx	localStorage crm_pending_ticket	Frontend
NFR-ST-15	Ticket archiving 365+ days	archiveService.js	archiveOldTickets() daily cron	POST /api/tickets/archive
NFR-SEC-04	Idle session timeout	AuthContext.jsx	30-min auto-logout	Frontend

4. Requirement to User Story Mapping

Maps functional and non-functional requirements to documented user stories (docs/user_stories.md).

Req ID	User Story	Description
FR-AUTH-01	US-01	Sales user can log in with email and password
FR-AUTH-02	US-01	Failed login returns 401 error
FR-AUTH-03	US-01	Unauthenticated request returns 401
FR-AUTH-04	US-02	Role-based access control enforcement
FR-LM-01	US-03	Sales rep creates a new lead
FR-LM-02	US-03	Duplicate email prevented at creation
FR-LM-03	US-04	Leads listed by priority score
FR-LM-04	US-05	Update lead pipeline stage
FR-LM-05	US-06	Delete a lead record
FR-LM-06	US-09	Add interaction notes to a lead
FR-SC-01	US-04	Priority score calculated 0-100
FR-SC-06	US-08	Admin assigns lead to sales rep
FR-SC-13	US-10	Campaign ID linked to lead
FR-ST-01	US-11	Support user creates a ticket
FR-ST-02	US-12	Update ticket status
FR-ST-04	US-09	Interaction logging for leads
FR-ST-11	US-13	Dashboard active leads KPI
FR-UC-12	US-13	Dashboard open tickets KPI
FR-ST-14	US-13	Dashboard monthly revenue KPI
FR-DOC-15	US-07	CSV export of leads
NFR-ST-05	US-12	Optimistic locking on tickets
NFR-ST-07	US-15, US-16, US-17, US-20	GDPR/KVKK masking and erasure
NFR-ST-12	US-13	Dashboard 2-second auto-refresh
NFR-ST-14	US-19	Offline ticket draft caching
NFR-ST-15	US-15	Ticket archiving (365+ days)
NFR-SEC-04	US-18	30-minute idle session timeout

5. Requirement to UI Component Traceability

Maps requirements to frontend React components and pages that implement or display them.

Req ID	UI Component	Description
FR-AUTH-01/02	LoginPage.jsx	Email/password form, JWT stored in localStorage
FR-AUTH-03/04	ProtectedLayout.jsx, RoleRoute.jsx	Route guards redirect to /login or show 403
FR-LM-01/02	LeadPage.jsx, LeadForm.jsx	Lead table + create/edit modal form
FR-LM-03	LeadPage.jsx	Table sorted by priority_score DESC
FR-LM-04/05	LeadPage.jsx	Edit/Delete buttons per row
FR-LM-06	LeadProfilePage.jsx	Interaction timeline with add-note form
FR-SC-01	LeadPage.jsx, LeadForm.jsx	Score displayed in table, calculated at creation
FR-SC-06	LeadProfilePage.jsx	Admin assignment dropdown
FR-ST-01/02	TicketPage.jsx, TicketForm.jsx	Ticket list + create/update forms
FR-ST-09	TicketPage.jsx	Ticket list table for support users
FR-ST-11, FR-UC-12, FR-ST-14	DashboardPage.jsx	4 KPI cards with 2s auto-refresh
FR-DOC-15	LeadPage.jsx	CSV Export button
NFR-ST-02	index.css	Responsive layout min-width 360px
NFR-ST-05	TicketPage.jsx	Conflict detection alert on stale update
NFR-ST-07	LeadProfilePage.jsx	PII masked for support role display
NFR-ST-12	DashboardPage.jsx	setInterval(fetchData, 2000)
NFR-ST-14	TicketPage.jsx	Offline draft banner with Retry/Dismiss
NFR-SEC-04	AuthContext.jsx	30-min idle auto-logout with activity listeners

6. Requirement to Test Case Traceability Matrix

Maps each functional/non-functional requirement to specific test cases across unit, functional, and BDD test suites. All 53 tests and 20 BDD scenarios pass.

Req ID	Test Case(s)	Test File(s)	Status
FR-AUTH-01	Unit: authService - generateToken returns signed JWT Functional: POST /api/auth/login returns 200 BDD: auth_login.feature - Successful login	authService.test.js api.test.js auth_login.feature	PASS
FR-AUTH-02	Unit: authService - comparePasswords returns false Functional: POST /api/auth/login returns 401 BDD: auth_login.feature - Failed login	authService.test.js api.test.js auth_login.feature	PASS
FR-AUTH-03	Functional: GET /api/leads returns 401 without token BDD: auth_login.feature - Rejected without token	api.test.js auth_login.feature	PASS
FR-AUTH-04	Functional: GET /api/leads returns 403 wrong role BDD: auth_login.feature - Rejected insufficient role	api.test.js auth_login.feature	PASS
FR-LM-01	Functional: POST /api/leads returns 201 BDD: lead_registration.feature - Successfully register	api.test.js lead_registration.feature	PASS
FR-LM-02	Functional: POST /api/leads duplicate returns 400 BDD: lead_registration.feature - Reject duplicate BDD: duplicate_email.feature - Reject existing email	api.test.js lead_registration.feature duplicate_email.feature	PASS
FR-LM-03	Functional: GET /api/leads sorted DESC BDD: lead_scoring.feature - Sorted by score	api.test.js lead_scoring.feature	PASS
FR-SC-01	Unit: scoringService - caps/rounds scores correctly (6 tests) BDD: lead_scoring.feature - High/Low score scenarios	scoringService.test.js lead_scoring.feature	PASS
FR-ST-01	BDD: ticket_creation.feature - Create support ticket	ticket_creation.feature	PASS
FR-ST-02	Unit: ticketService - updates status on valid updated_at BDD: ticket_creation.feature - Update ticket status	ticketService.test.js ticket_creation.feature	PASS
FR-ST-04	Unit: leadService - addLog creates note, getLogs returns logs Functional: POST /api/leads/:id/logs returns 201 Functional: GET /api/leads/:id/logs returns 200 BDD: interaction_logging.feature - Add note + Retrieve logs	leadService.test.js api.test.js interaction_logging.feature	PASS
FR-UC-05	Unit: leadService - accepts valid stages, rejects invalid Functional: PUT /api/leads/:id pipeline_stage returns 200 BDD: pipeline_stage.feature - Update to Qualified + Reject invalid	leadService.test.js api.test.js pipeline_stage.feature	PASS
FR-SC-06	Unit: leadService - updates user_id for assignment Unit: leadService - throws LEAD_NOT_FOUND	leadService.test.js	PASS
FR-ST-09	Functional: GET /api/tickets returns 200 with array	api.test.js	PASS
FR-ST-11	Unit: dashboardService - returns activeLeads count Functional: GET /api/dashboard returns all KPI fields BDD: dashboard_kpis.feature - All KPI fields	dashboardService.test.js api.test.js dashboard_kpis.feature	PASS
FR-UC-12	Unit: dashboardService - returns openTickets count Functional: GET /api/dashboard returns all KPI fields BDD: dashboard_kpis.feature - All KPI fields	dashboardService.test.js api.test.js dashboard_kpis.feature	PASS
FR-SC-13	Unit: leadService - stores campaign_id Functional: POST /api/leads stores campaign_id	leadService.test.js api.test.js	PASS
FR-ST-14	Unit: dashboardService - returns monthlyRevenue sum Functional: GET /api/dashboard returns all KPI fields BDD: dashboard_kpis.feature - All KPI fields	dashboardService.test.js api.test.js dashboard_kpis.feature	PASS
FR-DOC-15	Functional: GET /api/leads/export/csv returns text/csv BDD: csv_export.feature - Export leads as CSV	api.test.js csv_export.feature	PASS
NFR-ST-05	Unit: ticketService - CONFLICT on stale updated_at	ticketService.test.js	PASS
NFR-ST-07	Unit: leadService - erasePersonalData anonymises PII Unit: dashboardService - masks top5 PII for support Functional: GET /api/leads/:id - PII masked (support) Functional: GET /api/leads/:id - unmasked (sales)	leadService.test.js dashboardService.test.js api.test.js	PASS

Req ID	Test Case(s)	Test File(s)	Status
NFR-ST-10	Unit: scoringService - rounds to integer, <500ms	scoringService.test.js	PASS
NFR-SEC-01	Unit: authService - hashPassword returns bcrypt hash	authService.test.js	PASS
NFR-SEC-02	Functional: 401 without token	api.test.js	PASS
NFR-SEC-03	Functional: 403 for insufficient role	api.test.js	PASS
NFR-ST-12	Code review: DashboardPage setInterval(2000)	DashboardPage.jsx	PASS
NFR-ST-14	Code review: TicketPage localStorage crm_pending_ticket	TicketPage.jsx	PASS
NFR-ST-15	Code review: archiveService.js daily cron + ArchivedTicket table	archiveService.js	PASS
NFR-SEC-04	Code review: AuthContext.jsx 30-min idle timeout	AuthContext.jsx	PASS

7. Test Coverage Summary

Aggregated test coverage statistics across all three test tiers.

Test Tier	Count	Scope	Result
Unit Tests (Jest)	38	5 suites: authService(6), scoringService(6), ticketService(6), leadService(12), dashboardService(6)	38/38 PASS
Functional Tests (Supertest)	15	api.test.js: CRUD, auth, CSV, PII, pipeline, dashboard, interaction logs	15/15 PASS
BDD Scenarios (Cucumber)	20	9 feature files: auth(4), lead_reg(2), scoring(3), tickets(2), dup_email(2), interaction(2), pipeline(2), dashboard(2), csv(1)	20/20 PASS
Total	73	All three tiers cover unit logic, API integration, and acceptance criteria	73/73 PASS

Requirements with Automated Test Coverage

25 requirements have automated tests (unit, functional, or BDD): FR-AUTH-01, FR-AUTH-02, FR-AUTH-03, FR-AUTH-04, FR-LM-01, FR-LM-02, FR-LM-03, FR-SC-01, FR-SC-06, FR-SC-13, FR-ST-01, FR-ST-02, FR-ST-04, FR-ST-09, FR-ST-11, FR-UC-05, FR-UC-12, FR-ST-14, FR-DOC-15, NFR-ST-05, NFR-ST-07, NFR-ST-10, NFR-SEC-01, NFR-SEC-02, NFR-SEC-03

Requirements Verified by Code Review

4 requirements verified by code review and manual testing (frontend-only or infrastructure): NFR-ST-12, NFR-ST-14, NFR-ST-15, NFR-SEC-04

8. Full Cross-Reference Matrix

Complete traceability from requirement through user story, source code, test, and UI for every tracked requirement.

Req ID	User Story	Source Code	Test Coverage	UI Component
FR-AUTH-01	US-01	authController.js	authService.test.js, api.test.js, auth_login.feature	LoginPage.jsx
FR-AUTH-02	US-01	authController.js	authService.test.js, api.test.js, auth_login.feature	LoginPage.jsx
FR-AUTH-03	US-01	authMiddleware.js	api.test.js, auth_login.feature	ProtectedLayout.jsx
FR-AUTH-04	US-02	rbacMiddleware.js	api.test.js, auth_login.feature	RoleRoute.jsx
FR-LM-01	US-03	leadController.js	api.test.js, lead_registration.feature	LeadPage.jsx
FR-LM-02	US-03	leadController.js	api.test.js, lead_registration.feature, duplicate_email.feature	LeadPage.jsx
FR-LM-03	US-04	leadController.js	api.test.js, lead_scoring.feature	LeadPage.jsx
FR-LM-04	US-05	leadController.js	leadService.test.js	LeadPage.jsx
FR-LM-05	US-06	leadController.js	—	LeadPage.jsx
FR-LM-06	US-09	leadController.js	leadService.test.js, api.test.js, interaction_logging.feature	LeadProfilePage.jsx
FR-SC-01	US-04	scoringService.js	scoringService.test.js, lead_scoring.feature	LeadForm.jsx
FR-SC-06	US-08	leadController.js	leadService.test.js	LeadProfilePage.jsx
FR-SC-13	US-10	leadController.js	leadService.test.js, api.test.js	LeadPage.jsx
FR-ST-01	US-11	ticketController.js	ticket_creation.feature	TicketPage.jsx
FR-ST-02	US-12	ticketController.js	ticketService.test.js, ticket_creation.feature	TicketPage.jsx
FR-ST-04	US-09	leadController.js	leadService.test.js, api.test.js, interaction_logging.feature	LeadProfilePage.jsx
FR-ST-09	US-11	ticketController.js	api.test.js	TicketPage.jsx
FR-ST-11	US-13	dashboardController.js	dashboardService.test.js, api.test.js, dashboard_kpis.feature	DashboardPage.jsx
FR-UC-05	US-05	leadController.js	leadService.test.js, api.test.js, pipeline_stage.feature	LeadProfilePage.jsx
FR-UC-12	US-13	dashboardController.js	dashboardService.test.js, api.test.js, dashboard_kpis.feature	DashboardPage.jsx
FR-ST-14	US-13	dashboardController.js	dashboardService.test.js, api.test.js, dashboard_kpis.feature	DashboardPage.jsx
FR-DOC-15	US-07	leadController.js	api.test.js, csv_export.feature	LeadPage.jsx
NFR-ST-05	US-12	ticketController.js	ticketService.test.js	TicketPage.jsx
NFR-ST-07	US-16,17,20	leadController.js, userController.js	leadService.test.js, dashboardService.test.js, api.test.js	LeadProfilePage.jsx
NFR-ST-10	—	scoringService.js	scoringService.test.js	—
NFR-ST-12	US-13	DashboardPage.jsx	Code review	DashboardPage.jsx
NFR-ST-14	US-19	TicketPage.jsx	Code review	TicketPage.jsx
NFR-ST-15	US-15	archiveService.js	Code review	—

Req ID	User Story	Source Code	Test Coverage	UI Component
NFR-SEC-01	—	authService.js	authService.test.js	—
NFR-SEC-02	US-01	authMiddleware.js	api.test.js	—
NFR-SEC-03	US-02	rbacMiddleware.js	api.test.js	—
NFR-SEC-04	US-18	AuthContext.jsx	Code review	—